



Protein Chemistry & Haemoglobin

MBBS BIOCHEMISTRY · PAPER 1

1° - 4° Structure

 α -Helix · β -Sheet

Hb vs Mb

 O_2 Dissociation

2,3-BPG · Bohr Effect

HbS · HbF · HbA

Haem Degradation

Bilirubin Metabolism

Plasma Proteins

Electrophoresis

EXAM PRIORITY MAP

Prioritize your revision based on high-yield exam patterns. Focus heavily on Tier 1 and Tier 2 sections for last-day prep.

● TIER 1 (Critical High-Yield)

- Sec 01: Protein Structure Levels
- Sec 04: O_2 Dissociation Curve
- Sec 05: 2,3-BPG & Bohr Effect
- Sec 06: HbA · HbS · HbF
- Sec 08: Bilirubin & Jaundice

● TIER 2 (Medium-Yield)

- Sec 02: α -Helix & β -Sheet
- Sec 03: Hb vs Myoglobin
- Sec 10: Electrophoresis (SPEP)

● TIER 3 (Low-Yield / Collapsed)

- Sec 07: Haem Degradation
- Sec 09: Plasma Proteins

🔥 LAST DAY REVISION MODE

Speed-read the high-yield essentials! The module has been optimized for rapid revision:

- ♦ All sections tagged with ● **TIER 1**, ● **TIER 2**, or ● **TIER 3** priority badges.
- ♦ ● TIER 3 sections are automatically collapsed into interactive details tags to prevent cognitive overload.
- ♦ High-priority ⚠ **EXAM TRAPS** are injected at critical conceptual points to save you from common mistakes.
- ♦ Use standard **Ctrl+P** to print this entire module as an optimized high-contrast PDF with all sections automatically expanded.

01

Levels of Protein Structure

● TIER 1

⚡ 4 levels exist.

Primary → **Secondary** → **Tertiary** → **Quaternary**

. Each higher level depends on the one below it.

1°
Primary Structure
Peptide Bond (Covalent)
Linear sequence of amino acids. Determined by the gene. **Sanger's method** identifies it. Example: Insulin (51 AA)

2°
Secondary Structure
H-bonds (backbone NH...CO)

Regular repeating patterns: **α -helix**, **β -pleated sheet**, β -turns. Stabilised by hydrogen bonds between backbone atoms.

3°

Tertiary Structure

Hydrophobic + H-bond + Ionic Bond + Disulphide

3D folding of a single polypeptide. Hydrophobic core. Disulphide bonds (covalent). Determines biological function.

4°

Quaternary Structure

Non-covalent + some S-S

Assembly of 2+ polypeptide subunits. Example: **Haemoglobin ($\alpha_2\beta_2$)**. Not all proteins have this level.

● MNEMONIC – BONDS BY LEVEL

"Pretty Smart Tertiary Queens" → Peptide, Stable H-bonds, Three types (hydrophobic+H+ionic+S-S), Quaternary = all of the above

Or simpler: **1° Peptide** → **2° Hydrogen** → **3° Multiple (HISD: Hydrophobic, Ionic, S-S, Disulphide)** → **4° Same as 3° but between chains**

🔗 CLINICAL CORRELATION – WHY PRIMARY STRUCTURE MATTERS

A single amino acid substitution in primary structure → disease. In **Sickle Cell Anaemia**, Glu→Val at position 6 of β -globin chain creates HbS. This demonstrates that *primary structure determines function* — a core biochemistry principle.

FORCES STABILISING PROTEIN STRUCTURE

FORCE	TYPE	WHERE IT ACTS	DISRUPTED BY
Peptide bond	Covalent	1° (backbone)	Proteases
H-bond	Non-covalent	2°, 3°	Urea, heat
Hydrophobic interactions	Non-covalent	3°	Detergents, organic solvents
Ionic/electrostatic	Non-covalent	3°	pH change, salt
Disulphide bond (-S-S-)	Covalent	3° (surface/interior)	Reducing agents (β -ME, DTT)
Van der Waals	Non-covalent	3°, 4°	Weak, easily disrupted

⚠ **Denaturation** disrupts 2°, 3°, 4° structure but NOT 1° (peptide bonds remain intact). Renaturation is possible if 1° is intact.

⚠ **EXAM TRAP:** Denaturation disrupts 2°, 3°, and 4° structures by breaking non-covalent bonds (and disulphide bonds), but it **NEVER disrupts the 1° structure** (covalent peptide bonds). Always state this explicitly to secure full marks!

02

α -Helix & β -Pleated Sheet

TIER 2

A-HELIX

★ EXAMINER FAV

- ✦ Right-handed helix (most proteins)
- ✦ **3.6 amino acids** per turn
- ✦ Rise per residue: **1.5 Å**
- ✦ Pitch (rise per turn): **5.4 Å**
- ✦ H-bonds: between **NH of residue n** and **C=O of residue n-4**
- ✦ H-bonds run **parallel to helix axis**
- ✦ R-groups point **outward** from helix
- ✦ Disrupted by: Proline (no NH), Glycine (too flexible), charged R-groups

EXAMPLE PROTEINS

Keratin (hair, nails): nearly entirely α -helical, coiled-coil structure

Myoglobin: 8 α -helical segments (A-H)

B-PLEATED SHEET

PYQ 2017, 2018

- ✦ Extended, zigzag conformation
- ✦ Two types: **Parallel** and **Antiparallel**
- ✦ H-bonds run **perpendicular to chain direction**
- ✦ Antiparallel: stronger H-bonds (more stable)
- ✦ R-groups alternate: above and below sheet plane
- ✦ Rise per residue: **3.5 Å**

FEATURE	PARALLEL	ANTIPARALLEL
Chain direction	Same	Opposite
H-bond angle	Bent	Linear
Stability	Less	More

FIBROUS VS GLOBULAR PROTEINS

PROPERTY	FIBROUS (STRUCTURAL)	GLOBULAR (FUNCTIONAL)
Shape	Long, rod-like	Spherical/compact
Solubility	Insoluble in water	Soluble
Dominant 2° structure	α -helix (keratin) or β -sheet (fibroin)	Mix of helix, sheet, coil
Role	Structure (hair, tendons, silk)	Enzyme, transport, Hb, Ab
Example	Keratin, Collagen, Fibroin	Haemoglobin, Myoglobin, Lysozyme

03

Haemoglobin vs Myoglobin

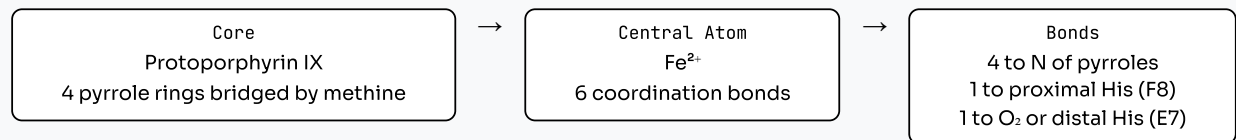
TIER 2

FEATURE	HAEMOGLOBIN (HB)	MYOGLOBIN (MB)
Location	RBCs (blood)	Muscle cells
Subunits	Tetramer: $\alpha_2\beta_2$ (HbA)	Monomer (1 chain)
MW	~68,000 Da	~17,000 Da
Haem groups	4 (one per subunit)	1
O ₂ affinity curve	Sigmoidal (S-shaped)	Hyperbolic
Cooperativity	Yes (allosteric)	No (no cooperativity)
Bohr effect	Yes (CO ₂ & H ⁺ ↓ affinity)	None
2,3-BPG effect	Yes (↓ O ₂ affinity)	None
Function	O ₂ transport	O ₂ storage
P ₅₀ (pO ₂ at 50% saturation)	~26 mmHg (higher = lower affinity)	~2 mmHg (lower = higher affinity)
T-state / R-state	T (tense/deoxy), R (relaxed/oxy)	N/A (monomer)
Secondary structure	8 α -helical segments per chain	8 α -helical segments (A–H)

HAEM GROUP STRUCTURE

Haem is a **porphyrin ring** (protoporphyrin IX) containing a central **Fe²⁺** ion.

► HAEM ARCHITECTURE

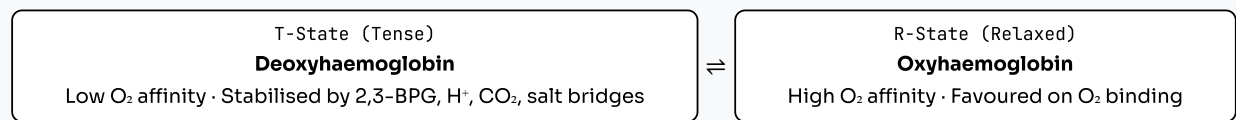


⚡ **Fe²⁺ (ferrous) binds O₂. Oxidation to Fe³⁺ (ferric) = methaemoglobin — cannot carry O₂.**

T-STATE VS R-STATE (ALLOSTERIC BEHAVIOUR)

★ HIGH YIELD

► HB CONFORMATIONAL SWITCH



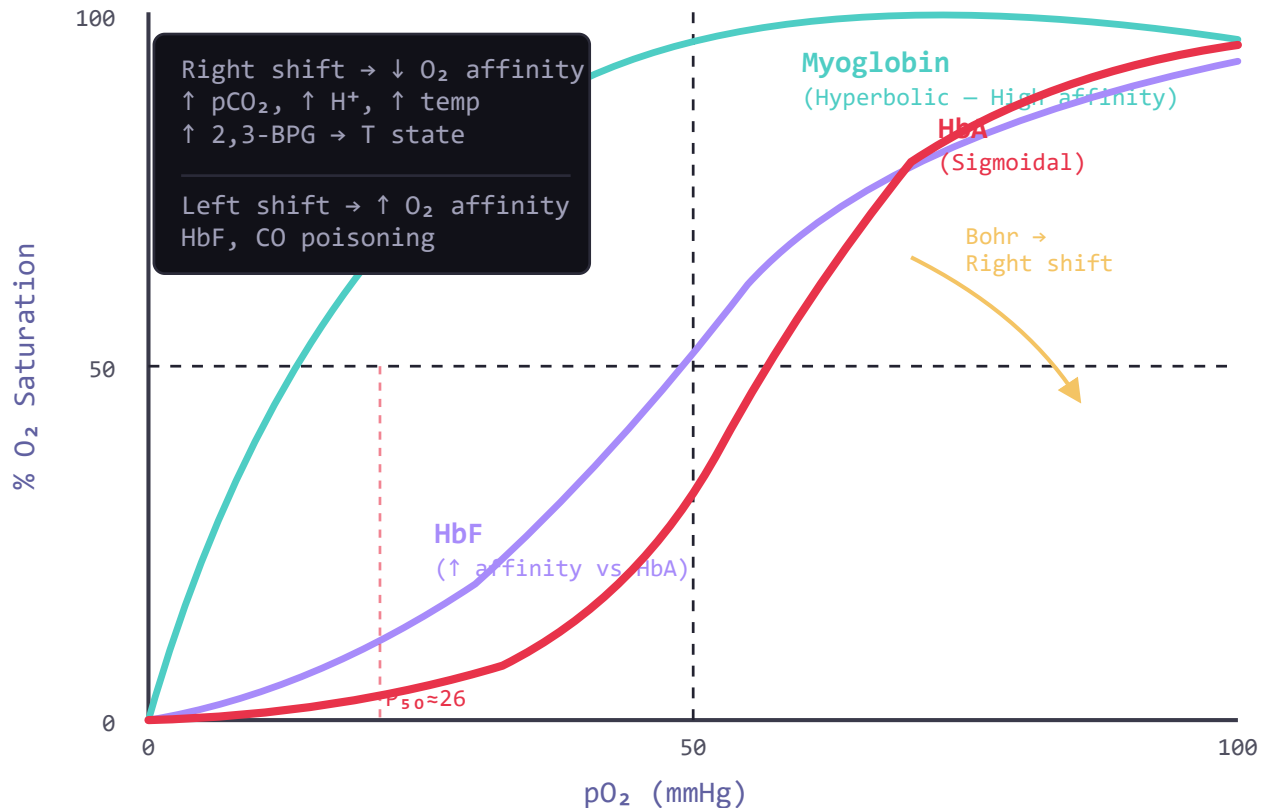
As first O₂ binds → conformational change → subsequent binding becomes EASIER → **cooperative binding** → Sigmoidal curve.

04

O₂ Dissociation Curve

TIER 1

▶ OXYGEN DISSOCIATION CURVES – HB VS MB VS HbF



RIGHT SHIFT → ↓ O₂ AFFINITY (O₂ UNLOADING)

- ◆ ↑ pCO₂ (exercise, tissues)
- ◆ ↑ H⁺ (acidosis) — Bohr effect
- ◆ ↑ Temperature
- ◆ ↑ 2,3-BPG

⚡ Right shift = **MORE O₂ delivered to tissues** . Good in exercise, fever.

LEFT SHIFT → ↑ O₂ AFFINITY (O₂ HOARDING)

- ◆ ↓ pCO₂ (hyperventilation)
- ◆ ↓ H⁺ (alkalosis)
- ◆ ↓ Temperature
- ◆ ↓ 2,3-BPG (stored blood)
- ◆ HbF (foetal Hb)
- ◆ CO poisoning

⚡ Left shift = **Hb grabs O₂ but won't release it** . Dangerous in CO poisoning.



EXAM TRAP: A **shift to the RIGHT** means **DECREASED oxygen affinity** (easier oxygen unloading, higher P₅₀). A shift to the **LEFT** means **INCREASED affinity** (harder unloading, lower P₅₀). Confusing these is the most common reason for losing marks!

05

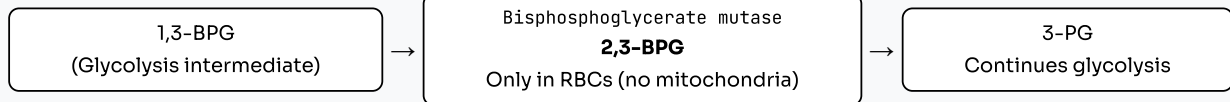
2,3-BPG & Bohr Effect

● TIER 1

2,3-BISPHOSPHOGLYCERATE (2,3-BPG)

★ PYQ EVERY YEAR

▶ RAPOPORT-LUEBERING CYCLE (HOW 2,3-BPG IS MADE)



Mechanism of Action:

- ◆ 2,3-BPG binds the **central cavity** of Hb between β -chains
- ◆ Only binds **T-state (deoxyHb)**
- ◆ Stabilises T-state $\rightarrow$ $\downarrow$ O_2 affinity $\rightarrow$ right shift
- ◆ **1 molecule** of 2,3-BPG per 1 Hb molecule

Clinical Significance:

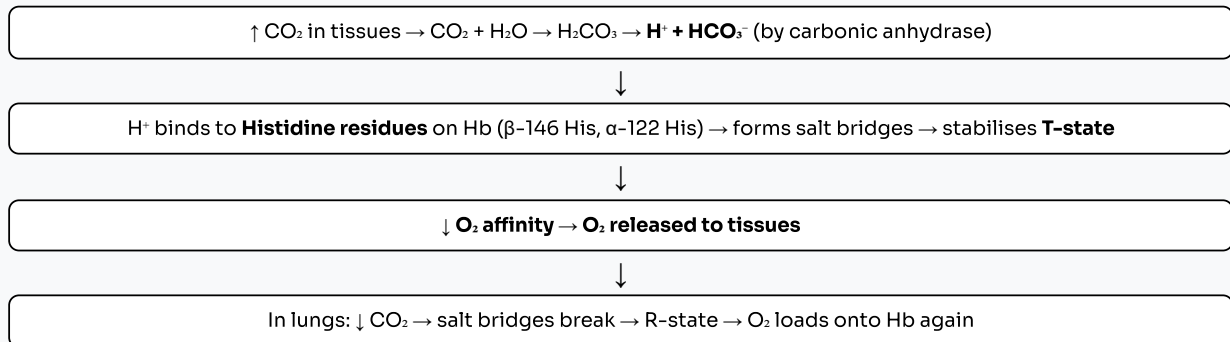
- ◆ **High altitude:** $\uparrow$ 2,3-BPG $\rightarrow$ $\uparrow$ O_2 delivery to tissues
- ◆ **Stored blood:** 2,3-BPG depletes $\rightarrow$ $\uparrow$ O_2 affinity $\rightarrow$ poor O_2 delivery (transfusion problem)
- ◆ **Anaemia:** $\uparrow$ 2,3-BPG (compensatory)
- ◆ **HbF:** doesn't bind 2,3-BPG (γ -chains) $\rightarrow$ $\uparrow$ affinity $\rightarrow$ foetal O_2 extraction from maternal Hb

BOHR EFFECT

PYQ 2013, MULTIPLE
★

- ⚡ Definition: The decrease in O_2 affinity of Hb caused by $\uparrow$ **CO_2 and/or H^+** . Facilitates O_2 delivery to metabolically active tissues.

▶ BOHR EFFECT MECHANISM



● KEY CONCEPT

Bohr = "H goes in $\rightarrow$ O_2 goes out" (tissues); " O_2 goes in $\rightarrow$ H goes out" (lungs)

06

HbA · HbS · HbF

TIER 1

HB TYPE	STRUCTURE	ELECTROPHORESIS	O ₂ AFFINITY	CLINICAL
HbA (Adult)	$\alpha_2\beta_2$	Moves fastest toward anode at pH 8.6	Normal (P_{50} 26 mmHg)	Normal adult haemoglobin
HbA ₂	$\alpha_2\delta_2$	Slow band	Similar to HbA	↑ in β -Thalassaemia (2–3%)
HbF (Foetal)	$\alpha_2\gamma_2$	Faster than HbS but slower than HbA	↑ (P_{50} ~20 mmHg)	↑ O ₂ affinity; normal until 6 months; ↑ in β -Thal, sickle cell
HbS (Sickle)	$\alpha_2\beta_2^S$	Moves toward cathode (different from HbA)	Normal when oxygenated	Glu→Val at β_6 ; polymerises when deoxygenated → sickling
HbM	Various α or β mutations at proximal/distal His	Abnormal	Low (Fe ³⁺ locked)	Methaemoglobinaemia
HbA _{1c} (Glycated)	HbA + glucose at β -N-terminal Val	Moves ahead of HbA	Slightly ↓	Diabetes monitoring (reflects 8–12 wk glucose)

SICKLE CELL ANAEMIA – MOLECULAR BASIS

★ PYQ REPEATEDLY

► HBS PATHOPHYSIOLOGY

Point mutation: **GAG** → **GTG** (codon 6, β -globin gene)

Amino acid change

Glutamate (charged, hydrophilic) → **Valine (non-polar, hydrophobic)**Creates a **"sticky patch"** on deoxyHbS surface
Val fits into complementary hydrophobic pocket on adjacent Hb molecule**Polymerisation** of deoxyHbS → long fibres → **Sickling** of RBCs

Haemolytic anaemia + Vaso-occlusion + Organ damage (spleen, kidney, brain)

🔗 WHY HbF IS PROTECTIVE IN SICKLE CELL

HbF ($\alpha_2\gamma_2$) does NOT polymerise with HbS. High HbF levels (as in neonates, or with hydroxyurea treatment) **inhibit sickling** by diluting HbS polymer formation. This is why symptoms appear only after HbF switches to HbA (~6 months of age).



EQ alert: "Patient with HbS often suffers from anaemia" — answer: haemolytic due to sickling + membrane damage. Sickled cells are rigid and get destroyed in spleen.

EQ TEMPLATE – HbF HAS MORE O₂ AFFINITY THAN HbA

"HbF has more affinity towards oxygen than HbA." [2019-S]

1. **Fact:** HbF has P₅₀ ~20 mmHg vs HbA P₅₀ ~26 mmHg → lower P₅₀ = higher O₂ affinity
2. **Structural Basis:** HbF has γ-chains instead of β-chains. γ-chains have Serine at position 143 instead of Histidine (β-143 His)
3. **2,3-BPG Interaction:** 2,3-BPG binds poorly to γ-chains (Ser 143 creates steric hindrance). Less 2,3-BPG effect → HbF stays in R-state more → ↑ O₂ affinity
4. **Physiological Significance:** Foetus extracts O₂ from maternal HbA across the placenta. HbF's higher affinity ensures O₂ transfer from maternal to foetal blood
5. **Clinical:** After birth, γ→β switch occurs; HbF → HbA by 6 months. Persistence of HbF (HPFH) is actually protective in sickle cell and β-thalassaemia



EXAM TRAP: In Sickle Cell Anaemia (HbS), the mutation is strictly a substitution of **Valine (hydrophobic)** for **Glutamic Acid (polar)** at **position 6 of the β-globin chain** (not the α-chain). Writing α-chain or wrong amino acids results in immediate zero for that section!

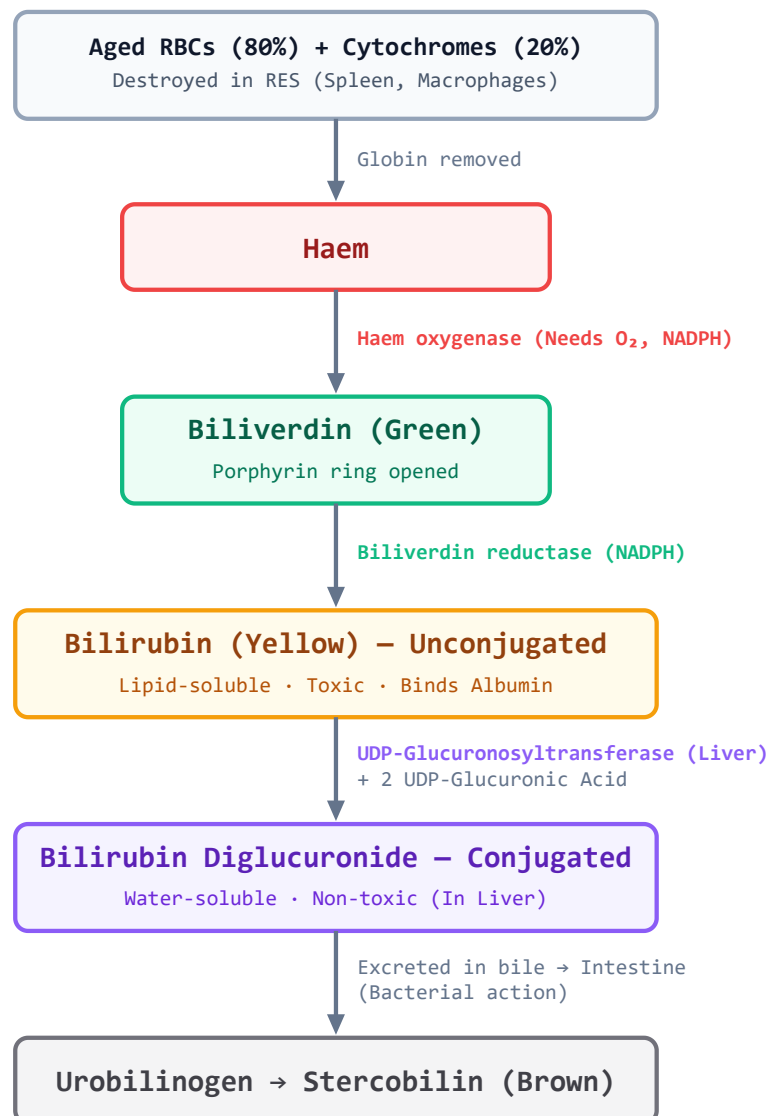
07

Haem Degradation

TIER 3

● TIER 3: Low-Yield Content — Click to expand Haem Degradation pathway

FIG 4.1: HAEM DEGRADATION PATHWAY



● MNEMONIC – HAEM BREAKDOWN COLOURS

"Haem Goes Bye-bye, Urine Stays" = Haem → Green (biliverdin) → Bilirubin (yellow) → Urobilinogen → Stercobilin

08 Bilirubin & Jaundice ● TIER 1

PARAMETER	HAEMOLYTIC (PRE-HEPATIC)	HEPATOCELLULAR	OBSTRUCTIVE (POST-HEPATIC)
Serum Unconjugated Bili	↑↑↑	↑	Normal/slightly ↑
Serum Conjugated Bili	Normal	↑	↑↑↑
Urine Bilirubin	Absent (indirect insoluble)	Present	Present ↑↑
Urine Urobilinogen	↑↑↑	↑	↓/Absent
Stool colour	Dark (↑ stercobilin)	Pale	Clay/pale (no bile)
Urine colour	Dark (urobilinogen)	Dark	Dark (bilirubin)
ALT/AST	Normal	↑↑↑	Normal/↑
Alkaline Phosphatase	Normal	↑	↑↑↑
Example	G6PD deficiency, Thalassaemia, RBC destruction	Hepatitis, Cirrhosis	Gallstones, Cholestasis, CA head of pancreas

NEONATAL/PHYSIOLOGICAL JAUNDICE PYQ 2011, 2017, 2021

Why it Occurs:

- ◆ HbF → HbA switch: massive RBC turnover
- ◆ Immature UGT1A1 in neonatal liver → poor conjugation
- ◆ Increased enterohepatic circulation (intestinal bacteria absent)
- ◆ Albumin less, so less binding → free bili crosses BBB → **Kernicterus**

Phototherapy Mechanism:

- ◆ Blue light (460 nm) → converts bilirubin (4Z,15Z) → photoisomers (4Z,15E lumirubin)
- ◆ Photoisomers are **water-soluble** → can be excreted in bile/urine WITHOUT conjugation
- ◆ Works without needing UGT1A1 → ideal for neonates
- ◆ Preferred over phenobarbitone in Crigler-Najjar Type I (no UGT at all)

VAN DEN BERGH REACTION PYQ 2019

Direct Reaction

- = Conjugated bilirubin reacts DIRECTLY with diazo reagent (no alcohol needed)
- = Water-soluble

Indirect Reaction

- = Unconjugated bilirubin reacts ONLY after adding alcohol (breaks albumin bond)
- = Lipid-soluble

⚡ **Total bilirubin = Direct + Indirect.** Normal: Total <1 mg/dL, Direct <0.3 mg/dL.

⚠ **EXAM TRAP:** Do not mix up the ****Van den Bergh reaction**** results! **Pre-hepatic (hemolytic) jaundice** gives an ****Indirect Positive**** (water-insoluble unconjugated bilirubin). **Post-hepatic (obstructive) jaundice** gives a ****Direct Positive**** (water-soluble conjugated bilirubin). **Hepatic jaundice** gives a ****Biphasic**** response.

09 Plasma *Proteins* TIER 3

● TIER 3: Low-Yield Content — Click to expand Plasma Proteins functions and properties

⚡ Normal Total Plasma Protein: **6–8 g/dL** . Albumin: **3.5–5 g/dL** . A:G ratio: **1.5–2.5:1**

PROTEIN	SITE OF SYNTHESIS	FUNCTION	CLINICAL SIGNIFICANCE
Albumin	Liver	Oncotic pressure, transport (bilirubin, FA, drugs, Ca ²⁺)	↓ in liver disease, nephrotic syndrome → oedema
α₁-Antitrypsin	Liver	Inhibits elastase (neutrophil)	Deficiency → emphysema, liver disease
α₂-Macroglobulin	Liver	Broad-spectrum protease inhibitor	↑ in nephrotic syndrome (too large to be lost)
Transferrin	Liver	Iron transport (Fe ³⁺ binding)	↓ in liver disease, malnutrition; ↑ in iron deficiency
Haptoglobin	Liver	Binds free Hb → prevents renal damage	↓ in haemolysis (consumed)
Ceruloplasmin	Liver	Copper transport (6–7 Cu atoms), ferroxidase activity	↓ in Wilson's disease ; Acute phase reactant
Fibrinogen	Liver	Coagulation (→ fibrin by thrombin)	↑ in inflammation; Acute phase reactant
Immunoglobulins	Plasma cells (B cells)	Antibody function (IgG, IgA, IgM, IgD, IgE)	↑ in infections; ↓ in immunodeficiency; M-spike in myeloma
Prothrombin	Liver (Vit K-dependent)	Clotting factor II (→ thrombin)	↓ in Vit K deficiency, liver disease
C-Reactive Protein (CRP)	Liver	Acute phase reactant; opsonin	↑ in infection, inflammation, MI

ACUTE PHASE REACTANTS PYQ 2014, 2022

Positive APR (↑ in inflammation):

- ◆ CRP, Serum Amyloid A
- ◆ Fibrinogen, Prothrombin
- ◆ Ceruloplasmin, Haptoglobin
- ◆ α₁-Antitrypsin, α₂-Macroglobulin
- ◆ Ferritin, Complement proteins

Negative APR (↓ in inflammation):

- ◆ **Albumin**
- ◆ Transferrin
- ◆ Transthyretin (Prealbumin)
- ◆ Retinol-binding protein

⚡ Negative APRs are liver "luxury proteins" — synthesis switches to positive APRs during stress.

EQ TEMPLATE – HYPOALBUMINAEMIA LEADS TO PERIPHERAL OEDEMA [2025]

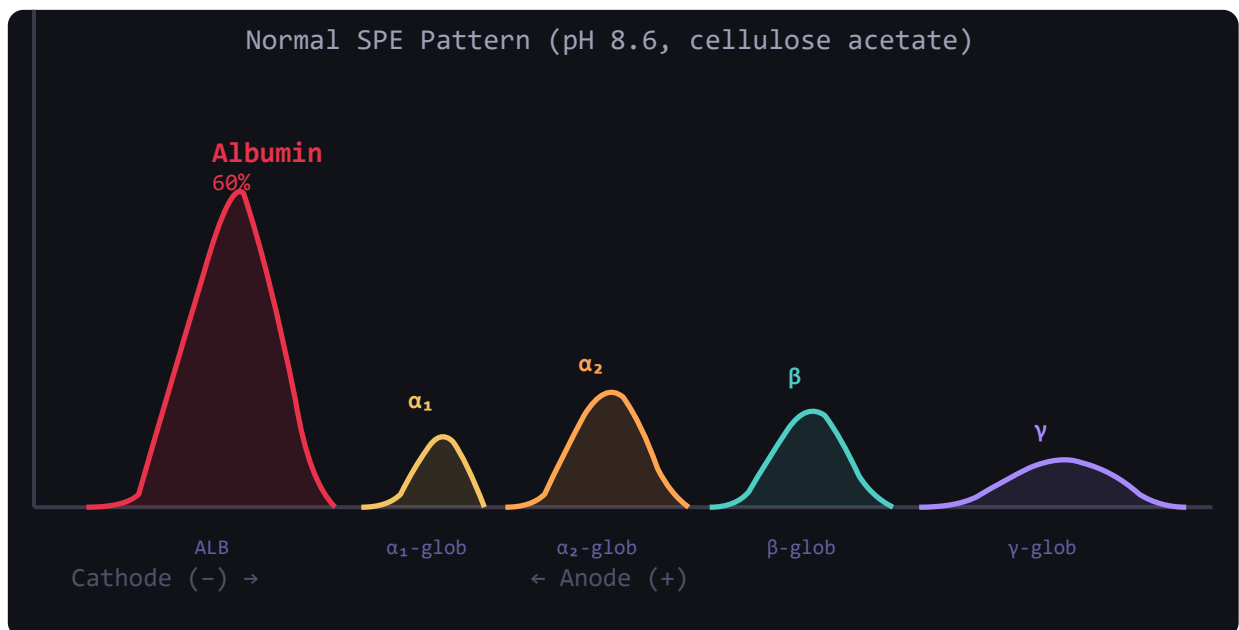
"Hypoalbuminaemia leads to peripheral oedema."

1. **Normal role of albumin:** Albumin (3.5–5 g/dL) is the principal contributor to **plasma oncotic (colloid osmotic) pressure** (~25 mmHg)
2. **Oncotic pressure function:** Draws water BACK into capillaries from interstitial space (opposing hydrostatic pressure that pushes fluid out)
3. **Hypoalbuminaemia:** ↓ plasma oncotic pressure → hydrostatic pressure > oncotic pressure → net fluid filtration into interstitium
4. **Result:** Peripheral oedema (pitting), ascites, pulmonary oedema, periorbital oedema
5. **Causes:** Liver disease (↓ synthesis), Nephrotic syndrome (↑ loss in urine), Malnutrition (kwashiorkor), Malabsorption
6. **Example:** Nephrotic syndrome — serum albumin <2.5 g/dL → massive generalised oedema (anasarca)

10 Serum Protein *Electrophoresis* ● TIER 2

⚡ At pH 8.6 (alkaline), all serum proteins are **negatively charged** → migrate toward anode (+ve). Separation based on size and charge.

► SERUM PROTEIN ELECTROPHORESIS PATTERN – NORMAL VS PATHOLOGICAL



BAND	PROTEINS INCLUDED	NORMAL %	↑ IN	↓ IN
Albumin	Albumin only	55–65%	Dehydration	Liver disease, nephrotic, malnutrition, inflammation
α₁-globulin	α ₁ -Antitrypsin, α ₁ -glycoprotein	2–4%	Acute inflammation, pregnancy	α ₁ -Antitrypsin deficiency
α₂-globulin	Haptoglobin, Ceruloplasmin, α ₂ -Macroglobulin	6–12%	Nephrotic syndrome (↑ α ₂ -Macro)	Haemolysis (↓ haptoglobin)
β-globulin	Transferrin, β-Lipoprotein, Complement (C3, C4)	8–14%	Iron deficiency (↑ transferrin), Obstructive jaundice	Liver failure
γ-globulin	IgG (predominant), IgA, IgM, IgD, IgE	12–22%	Infections, autoimmune diseases; M-spike in myeloma	Immunodeficiency, nephrotic syndrome

Pathological SPE Patterns

Multiple Myeloma: Narrow, tall "M-spike" (monoclonal Ig) in γ or β region. ↓ normal Igs (immunoparesis).

Cirrhosis: ↓ Albumin + β-γ bridge (IgA bridges β and γ zones). ↑ γ polyclonal.

Nephrotic Syndrome: ↓↓ Albumin + ↑ α₂-globulin (α₂-Macroglobulin too big to be lost through GBM) + ↓ γ.

Acute Inflammation: ↓ Albumin + ↑ α₁ + ↑ α₂ (acute phase reactants). Fibrinogen absent in serum but present in plasma.

11 **Viva & Exam Q&A** ● TIER 1 PROTEIN STRUCTURE VIVAS**What bonds stabilise quaternary structure of proteins?** GROUP A

Quaternary structure is stabilised by the same **non-covalent forces** that stabilise tertiary structure between adjacent polypeptide chains:

- **Hydrophobic interactions** (predominant at subunit interfaces)
- **Hydrogen bonds** between chains
- **Ionic/electrostatic interactions** (salt bridges)
- **Disulphide bonds** (covalent, e.g., between chains of IgG)

Key example: Haemoglobin ($\alpha_2\beta_2$) — subunits held mainly by hydrophobic contacts and 8 ion pairs (salt bridges) in T-state.

Why can proline not be part of an α -helix? 2014

Proline has a **cyclic pyrrolidine ring** that:

- Has **no NH group** in its peptide bond → cannot donate H-bond → breaks helix H-bonding pattern
- Introduces a **rigid kink/bend** (restricted rotation around N-C α bond)
- Creates steric clash with adjacent residues

Proline is called a "helix breaker." It frequently occurs at bends and turns. Collagen is unusual — it forms a triple helix ONLY because of its Gly-Pro-X repeat.

Define denaturation. Which bonds are broken? Core Concept

Denaturation = Disruption of the native 3D structure of a protein without breaking **peptide bonds** (1° structure intact).

Bonds broken: H-bonds, hydrophobic interactions, ionic bonds, sometimes disulphide bonds.

Result: Loss of biological activity, altered solubility, viscosity changes.

Agents: Heat, urea/guanidinium (H-bonds), detergents (hydrophobic), acid/alkali (ionic), reducing agents (S-S bonds, e.g., β -mercaptoethanol).

Renaturation is possible if denaturing agent is removed and 1° structure is intact (e.g., ribonuclease A experiment by Anfinsen).

HAEMOGLOBIN VIVAS

Why is the O₂ dissociation curve of Hb sigmoidal while Mb is hyperbolic? PYQ 2014, 2016, 2023

Myoglobin: monomer with 1 haem. No cooperativity. Each molecule binds O₂ independently → **simple hyperbolic curve**. Very high affinity at all pO₂.

Haemoglobin: tetramer (α₂β₂). Exhibits **positive cooperativity**:

- First O₂ binds with LOW affinity (T-state)
- Triggers conformational change → R-state → second O₂ binds more easily
- Each successive binding is easier → **sigmoidal (S-shaped) curve**

Physiological significance: Sigmoid shape means Hb is very efficient — loads O₂ fully in lungs (high pO₂), unloads a large fraction in tissues (low pO₂). Mb's hyperbolic curve means it stays saturated throughout → ideal for O₂ storage, not transport.

What is Haemoglobin A1c? Why is it important in diabetes? PYQ 2011, 2018, 2016

HbA1c (Glycated Haemoglobin): Hb with glucose non-enzymatically (spontaneously) attached to the **N-terminal valine** of both β-chains via Schiff base → rearranges to stable ketoamine (Amadori product).

Why useful in diabetes:

- RBCs live ~120 days → HbA1c reflects **average blood glucose over 8–12 weeks**
- Unlike fasting glucose (snapshot), HbA1c = **cumulative glycaemic exposure**
- Normal: <5.7%; Pre-diabetes: 5.7–6.4%; DM: ≥6.5%
- Target in treated DM: <7% (to prevent microvascular complications)

Not affected by short-term glucose fluctuations or time of day. Limitations: haemolytic anaemia (falsely low), sickle cell trait may affect accuracy.

What is methaemoglobinaemia? How does it occur? Core

Methaemoglobin (MetHb): Hb in which iron is oxidised from **Fe²⁺ → Fe³⁺**. Fe³⁺ cannot reversibly bind O₂.

Causes: Oxidising drugs/toxins (dapsone, nitrites, sulphonamides, prilocaine), Congenital HbM (proximal/distal His mutations), NADH-metHb reductase deficiency.

Effect: ↓ functional Hb + left shift of remaining Hb (cannot release O₂) → tissue hypoxia.

Clinical: Cyanosis with **chocolate brown blood**. SpO₂ falsely reads ~85% by pulse oximetry.

Treatment: Methylene blue (activates NADPH-metHb reductase via G6PD) OR Vitamin C (non-enzymatic reduction).

 BILIRUBIN & PLASMA PROTEIN VIVAS

Why does conjugated bilirubin appear in urine but unconjugated does not? PYQ 2019

Unconjugated bilirubin:

- Lipid-soluble, tightly bound to albumin in blood
- Too large as albumin complex → NOT filtered by glomerulus
- Even if some is free, it binds to tubular cells (not excreted)
- Therefore: **absent in urine even when serum levels high** (haemolytic jaundice — "acholouric jaundice")

Conjugated (direct) bilirubin:

- Water-soluble (2 glucuronic acids attached)
- NOT bound to albumin → freely filterable at glomerulus
- When serum levels rise → appears in urine (bilirubinuria)
- Seen in: hepatocellular jaundice, obstructive jaundice

What are chaperones? What is their role in protein folding? PYQ 2016, 2018

Chaperones are proteins that assist in the correct folding of newly synthesised polypeptides. They do NOT contribute to the final structure (not part of the final protein).

Mechanism:

- Bind to exposed **hydrophobic patches** on unfolded/partially folded proteins
- Prevent premature or incorrect interactions (misfolding, aggregation)
- Provide a protected environment for folding (e.g., GroEL/GroES barrel in bacteria, HSP60 in mitochondria)
- ATP-dependent (most chaperones use ATP hydrolysis)

Types: Hsp70 (cytosol, early folding), Hsp90 (steroid receptor stabilisation), Hsp60 (GroEL — barrel type), Calnexin/Calreticulin (ER, glycoprotein folding).

Misfolding despite chaperones → ubiquitin-proteasome degradation of aberrant proteins.
Accumulation of misfolded proteins → Prion diseases, Alzheimer's, Parkinson's.

What are Bence Jones proteins? Significance? Myeloma

Bence Jones proteins: Free immunoglobulin **light chains** (κ or λ) produced in excess by myeloma plasma cells and excreted in urine.

Detection: Heat precipitation at 40–60°C, redissolve at 100°C (classic test). Modern: urine protein electrophoresis or immunofixation.

Significance: Diagnostic of **multiple myeloma**, Waldenström macroglobulinaemia, or light chain disease. Cause renal tubular damage (cast nephropathy = "myeloma kidney").

12 Essay Question *Templates* ● TIER 1

 EQ – 2,3-BPG HELPS IN DELIVERY OF OXYGEN TO TISSUES [PYQ 2011]

"2,3-BPG helps in delivery of oxygen to the tissues."

1. **Source:** Made from 1,3-BPG in RBCs via Rapoport-Luebering cycle (bisphosphoglycerate mutase). Concentration in RBCs ~4–5 mmol/L (equimolar with Hb)
2. **Binding Site:** Binds the central cavity of deoxyHb between the β -chains (involves Lys82, His143, His2, Val1 of each β -chain). One molecule per Hb tetramer
3. **Mechanism:** 2,3-BPG stabilises the **T-state (deoxyHb)** by cross-linking the β -chains with multiple ionic bonds $\rightarrow$ $\downarrow$ O_2 affinity $\rightarrow$ right shift of ODC
4. **Net Effect:** At tissue level (low pO_2), Hb releases MORE O_2 than it would without 2,3-BPG $\rightarrow$ better O_2 unloading
5. **Regulatory contexts:** $\uparrow$ at high altitude (compensatory), in anaemia, in hypoxia. $\downarrow$ in stored blood (2,3-DPG depletes $\rightarrow$ less O_2 delivery $\rightarrow$ risk in massive transfusion)
6. **HbF comparison:** HbF (γ -chains) binds 2,3-BPG poorly $\rightarrow$ higher O_2 affinity $\rightarrow$ ideal for transplacental O_2 transfer

 EQ – PATIENT WITH HBS OFTEN SUFFERS FROM ANAEMIA [PYQ 2013, 2017]

"Patient with Hb-S often suffers from anaemia."

1. **Molecular defect:** GAG $\rightarrow$ GTG mutation in β -globin gene, codon 6. Glu (polar) $\rightarrow$ Val (non-polar). Creates "sticky patch" on deoxyHbS surface
2. **Polymerisation:** In low- O_2 conditions, deoxyHbS fibres polymerise $\rightarrow$ RBCs become sickle-shaped, rigid, non-deformable
3. **Haemolysis:** Sickled RBCs are rigid $\rightarrow$ cannot pass through spleen $\rightarrow$ destroyed by splenic macrophages (extravascular haemolysis). Also intravascular haemolysis due to membrane damage. RBC survival $\downarrow$ from 120 days $\rightarrow$ 20–25 days
4. **Vaso-occlusion:** Sickled cells block microvasculature $\rightarrow$ ischaemic pain crises, organ infarction (spleen, kidney, brain, bones)
5. **Type of anaemia:** Normocytic normochromic haemolytic anaemia with compensatory reticulocytosis
6. **Complications:** Splenomegaly (early), autosplenectomy (late), aplastic crisis (parvovirus B19), susceptibility to infections (encapsulated organisms)

EQ - MYOGLOBIN DOES NOT EXHIBIT BOHR EFFECT [PYQ 2015]

"Myoglobin does not exhibit Bohr effect."

- Bohr effect definition:** The decrease in O₂ affinity of Hb when pH drops ($\uparrow$ H⁺) or pCO₂ rises, mediated via salt-bridge formation in T-state Hb
- Structural basis of Bohr in Hb:** H⁺ protonates histidine residues (β -146 His, α -122 His) $\rightarrow$ salt bridges form $\rightarrow$ stabilises T-state $\rightarrow$ $\downarrow$ O₂ affinity
- Why Mb shows no Bohr effect:** Myoglobin is a **monomer** — has no T-state/R-state equilibrium, no cooperative binding, no allosteric subunit communication. The key histidine residues involved in Bohr salt-bridges in Hb are not involved in the same way in monomeric Mb
- Physiological implication:** Mb's function is O₂ STORAGE in muscle. It should NOT release O₂ until pO₂ is very low. Hence, being unresponsive to pH/CO₂ changes is appropriate
- Conclusion:** Bohr effect is exclusively an allosteric property of *oligomeric haemoglobin*, made possible by subunit cooperativity — absent in the monomer myoglobin

MASTER MNEMONICS - MODULE 04

SPE BANDS IN ORDER

A-AlphaA-AlphaB-Beta-Gamma
"Angry Aardvarks Always Battle Gorillas"

Albumin $\rightarrow \alpha_1 \rightarrow \alpha_2 \rightarrow \beta \rightarrow \gamma$ (anode to cathode)

RIGHT SHIFT CAUSES (BOHR)

"CADET face RIGHT"
CO₂ $\uparrow$, Acid $\uparrow$, DPG $\uparrow$, Exercise, Temp $\uparrow$

PROTEIN STRUCTURE BONDS

1°=Peptide, 2°=H, 3°=HISD
(H=H-bond, I=Ionic, S=Sulphide, D=van Der waals)

HAEM DEGRADATION PRODUCTS

"Haem Goes Bright Yellow Under Stress"
H $\rightarrow$ Biliverdin $\rightarrow$ Bilirubin $\rightarrow$ Urobilinogen $\rightarrow$ Stercobilin

EXAM DAY HIGH-YIELD QUICK FIRE

TOPIC	THE ANSWER EXAMINER WANTS
Sickle cell mutation	GAG→GTG Glu(6)→Val β -globin codon 6
Why HbF has $\uparrow$ O₂ affinity	γ -chains bind 2,3-BPG poorly (Ser143 vs His143)
Bohr effect key residue	β -146 Histidine
2,3-BPG binding site	Central cavity between 2 β -chains (T-state only)
Direct vs Indirect bilirubin test	Van den Bergh — Direct reacts without alcohol; Indirect needs alcohol
Neonatal jaundice phototherapy	460 nm blue light converts bili to water-soluble lumirubin
Haptoglobin function	Binds free Hb in plasma → prevents renal damage → $\downarrow$ in haemolysis
M-spike on electrophoresis	Multiple myeloma — monoclonal Ig spike in γ region
Ceruloplasmin $\downarrow$	Wilson's disease (Cu accumulates in liver, brain, eye: Kayser-Fleischer rings)
α-helix residues per turn	3.6 residues; 1.5Å rise/residue; 5.4Å pitch
Prion disease mechanism	Misfolding of PrP ^c → PrP ^{sc} (β -sheet) → aggregation
Rate limiting step of haem catabolism	Haem oxygenase (first step)

MBBS Biochemistry – Module 04: Protein Chemistry & Haemoglobin
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